Iceberg Tables

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Section Introduction-Iceberg Tables Topics

- What is Apache Iceberg ?
- Key features of Apache Iceberg.
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- What are Snowflake Iceberg tables ?
- What is a Data Catalog and why do we care ?
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- What is the difference between External tables and Iceberg tables
- Summary.

What is Apache Iceberg ?

Apache Iceberg is an open-source high performance open table format for huge analytic tables.

With Apache Iceberg you can create tables inside the data lake on AWS that can be treated just like regular tables and they support insert, update, delete and truncate , in addition to rollback and time travel.

Iceberg tables created on AWS can be read by engines like Spark, Presto, Hive and Impala.

Iceberg was started at Netflix by Ryan Blue and Dan Weeks to over come the limitations of Hive.

Iceberg format is supported by several vendors in addition to Snowflake like Cloudera, Dremio, Starburst and Tabular.

Key features of Apache Iceberg?

- Hidden Partitioning: Iceberg handles partitioning behind the scenes, simplifying data management and optimizing query performance.
- Snapshot Isolation: Ensures data consistency and integrity, allowing concurrent reads and writes.
- Schema Evolution: Supports adding, renaming, deleting and re-ordering fields without rewriting the entire dataset.
- Time Travel: Allows querying of data snapshots at specific point in time(Time Travel), enables rollback and facilitates data audit.
- File Format Agnostic: Works with popular file formats like Parquet, Avro and ORC
- Efficient Storage Management: Minimizes metadata size and improves read/write performance, especially for big datasets.

What are Snowflake Iceberg tables ?

Snowflake Iceberg tables are tables that are created inside Snowflake and reference table data stored in AWS Iceberg tables. They are similar to External tables with several differences.

Types of Snowflake tables

- Unmanaged Snowflake Iceberg tables.
- Managed Snowflake Iceberg tables.

Features and differences between them

- The syntax for creating unmanaged and managed tables are different.
- Unmanaged Snowflake Iceberg tables are read-only from Snowflake.
- Managed Snowflake Iceberg tables are read-write from Snowflake.
- Managed Snowflake Iceberg tables have twice the performance of unmanaged tables.
- Managed Snowflake Iceberg tables are created first inside Snowflake & data is stored on cloud storage when data is inserted into them.

What are Snowflake Iceberg tables ?

- Unmanaged Snowflake Iceberg tables require an AWS Iceberg table to be available to reference them.
- Unmanaged Snowflake Iceberg tables use External AWS Glue Catalog.
- Managed Snowflake Iceberg tables use Snowflake Catalog.

What is a data catalog and why do we care?

Data catalog is a record of the metadata of Iceberg tables created.

It stores information on the

- Columns of the table.
- Data types of the columns.
- Files and path associated with the table.
- Partitioning information if any.

Catalog information is required for Snowflake unmanaged tables to be able to reference the AWS Iceberg table. As the Iceberg table is inside AWS the catalog information is stored in GLUE which is the data catalog for AWS.

However if the table was a Snowflake managed Iceberg table the catalog information is stored inside Snowflake. This is important to remember because we need to provide catalog information when creating Iceberg tables in Snowflake. This catalog information is used to locate and retrieve data associated with the Iceberg table.

What are the difference and similarities between External table and Iceberg tables ?

External Table	Iceberg Table
External tables reference data stored in cloud storage.	Iceberg tables reference data stored in cloud storage.
External tables support formats like parquet ,ORC,JSON, Avro, CSV and XML	Iceberg tables do not support CSV,JSON and XML format, they do support parquet, ORC and AVRO.
External tables when partitioned need to have a directory structure created by the data engineer and files placed inside of them	Iceberg tables create the necessary partitions if we provide it with the partition column name.
External tables can use auto-refresh to automatically refresh the External table and register files with it.	Iceberg tables need to manually refresh(ALTER ...REFRESH) for Snowflake Unmanaged tables.
External tables can use auto-refresh to automatically refresh the External table and register files with it.	Managed Snowflake Iceberg tables do not need any manual refreshes as catalog is Snowflake.

What are the difference and similarities between External table and Iceberg tables ?

External Table	Iceberg Table
External tables are much slower as compared to iceberg tables	Managed Iceberg tables select performance is very high and close to Snowflake regular tables
External tables are much slower as compared to iceberg tables	Unmanaged Iceberg tables select performance is very high and 2x that of External tables.

Summary

- Apace Iceberg is a high-performance open table format for huge analytic tables and supports insert, update ,delete, truncate in addition too rollback and time travel.
- Iceberg tables created on AWS can be read by engines like Spark, Presto, Hive and Impala.
- Key features of Iceberg tables include hidden partitioning, snapshot isolation, schema evolution, time travel, supports several file formats and is efficient with storage management.
- Snowflake supports unmanaged and managed Iceberg tables.
- Unmanaged Snowflake Iceberg tables require an AWS Iceberg table to be available to reference them from Snowflake.
- Managed Snowflake Iceberg tables are created first inside Snowflake & data is stored on cloud storage when data is inserted into them in Snowflake.
- Unmanaged Snowflake Iceberg tables are read-only from Snowflake.
- Unmanaged Snowflake Iceberg tables use AWS Glue Data Catalog.
- Managed Snowflake Iceberg tables are read-write from Snowflake.
- Managed Snowflake Iceberg tables use Snowflake Catalog.

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